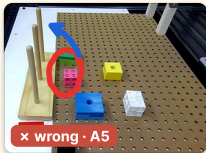
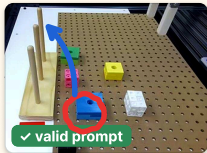


1 Drawn instruction $P = (I, C, A, L)$

put the **blue block** on the **peg**

↓ The target and direction is done by the sketch

👤 "put **this** **there**"



2 DrawVerify pre-execution trust gate



Qwen3-VL-4B + LoRA

sees only the marked image + caption

C

which

A

where

L

verb

X

agree

S

unique

3 Trained from simulator ground truth

pretrain

fine-tune



ground truth



guarded sketch



1 of 15 failures



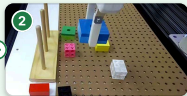
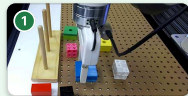
few human sketches

valid?

yes

no

✓ ACCEPT → execute



✗ REJECT(A5) → redraw



"please redraw"

Nothing runs. The user redraws and the gate checks the new sketch.

↺ redraw — the gate runs again on the new sketch